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Program/Project/Study Title: CUMULATIVE EFFECTS ASSESSMENT USING BAYESIAN NETWORK MODELLING: A MANAGEMENT PROSPECTIVE FOR ASIAN WETLANDS	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:							
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
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(7) Sector/Commodity

ENVIRONMENT/WETLANDS

(8) Discipline/Field of Specialization

ENVI.
SCI/INFORMATION
TECHNOLOGY

(9A) Abstract

The research aligns methodology of the Cumulative Effects Assessment (CEA) with Millennium Ecosystem Assessment (MA). The alignment was through integration of the concept of “human well-being”, defined as “basic material for good life, health, security, good social relations and freedom and choice”, into the CEA tool to provide an insight on the extent of cumulative effects on the multifunction of wetlands which consequently decreased provision of human well-being and on how effective management strategy can be done to balance development projects, wetlands integrity and human well-being. For an effective management intervention, cumulative effects assessment of the interactions between stressors and wetland valued ecosystem components (VECs) which in turn affects the delivery of wetland services (functions) are important to understand, hence, this study aims to develop a probability model of the cumulative effects of stressors to the VECs and services of wetlands. Assessment was done through: 1) secondary data collection on VECs and services offered by coastal wetlands in Southeast Asia; 2) elicitation of expert opinion, and; 3) analysis using Bayesian Network (Bayes Net) modelling. Bayes Net classification technique developed a probability model to determine the cumulative effects of stressors to VECs and wetland multi-function. Directed acyclic graph was generated to present correlations among stressors and conditional probability table (CPT) shows the probability distribution for every node in the graph. Multi-factorial analysis intrinsic to CEA is seen as a relevant approach to the wetland multi-function. The developed CEA methodological approach overcomes the difficulty of quantifying cumulative effects in which an enormous data complexity on interactions of numerous stressors and multiple functions existed. With an additional work further undertaken, this approach will be simplified making it possible to finalize the development of the tool straightforward enough which will facilitate sustainable multi-functional use of wetland resources through proper management of the existing stressors.

(9B) Keywords

Cumulative Effect Assessment (CEA), Bayesian Network Model, wetland

(10) Introduction

Wetlands are part of our natural ecosystems and have been vital to the development and survival of human communities. Wetland functions which include the storage of water, transformation of nutrients, growth of living matter, and diversity of wetland plants are very important to recognize. With this, ecological integrity of wetlands which continue on degrading has to be saved and improved for human well-being. For an effective management intervention, cumulative effects assessment of the interactions between stressors and wetland valued ecosystem components (VECs) which in turn affects the delivery of wetland functions are important to understand, hence, this study aims to develop a probability model of the cumulative effects of stressors to the VECs and functions of wetlands.

This research aligns methodology of the Cumulative Effects Assessment (CEA) with Millennium Ecosystem Assessment (MA). The alignment was through integration of the concept of “human well-being”, defined as “basic material for good life, health, security, good social relations and freedom and choice”, into the CEA tool to provide an insight on the extent of cumulative effects on the multifunction of wetlands which consequently decreased

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provision of human well-being and on how effective management strategy can be done to balance development projects, wetlands integrity and human well-being.

Cumulative effects assessment of multifunctional wetlands could be done through data mining. Data mining is an interdisciplinary field bringing together techniques from machine learning, pattern recognition, statistics, databases, visualization, and neural networks [6]. With data mining techniques, large amount of data in a database are collected, search through, and analyze to look for hidden patterns in a group of data that can be used to predict future behavior and trends. Data mining functionalities includes characterization, discrimination, association, classification, clustering, outlier and trend analysis, etc. [2]

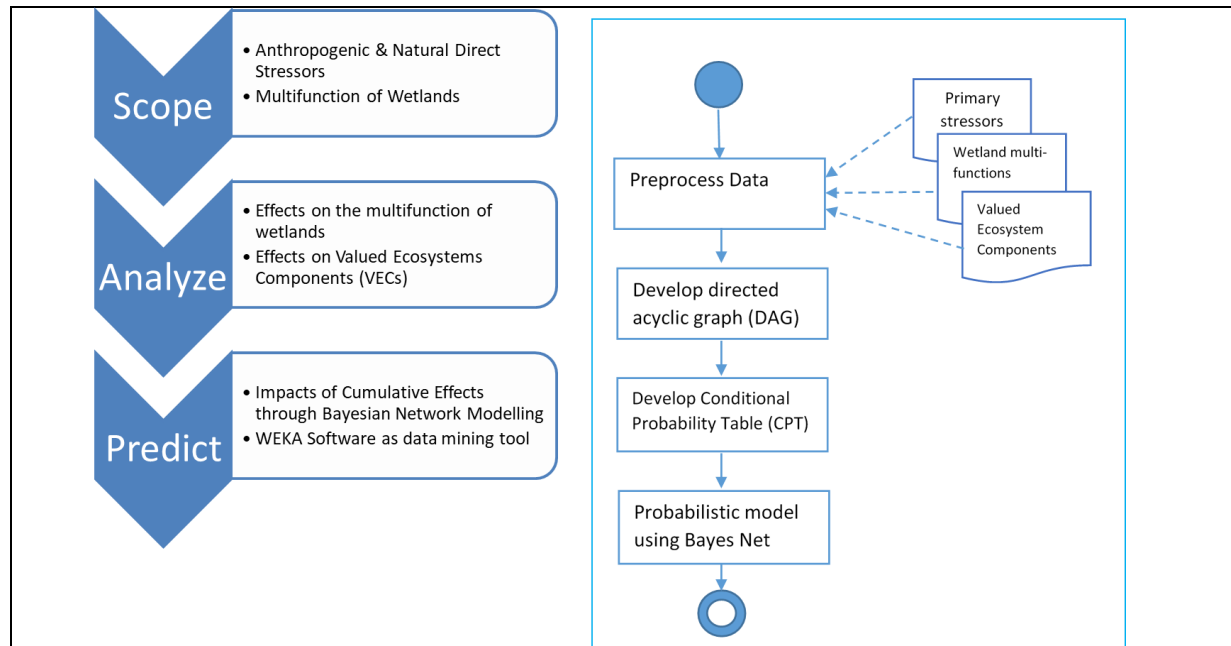
Classification data analysis is used to extract models to predict categorical class labels. The basic techniques for data classification includes decision tree classifiers, Bayesian classifiers, and rule-based classifiers. [2]

The objective of this study is to develop a probability model of the cumulative effects of drivers (stressors) to the multifunction (services) of wetlands.

(11) Materials and Methods or Methodology

Data that was used in this study was obtained from the existing data gathered from different experts in the Southeast Asia. Scoping of different primary direct drivers or stressors, valued ecosystem components and multifunction of wetlands were done. The primary direct drivers were: pollution, land use change, invasive species, over-exploitation, climate change, mining, commercial industries and agriculture. The identified valued ecosystem components were surface water, soil, aquatic resource, vegetation, biodiversity and resources used or obtained by community people aside from aquatic resources. Four general functions of coastal wetlands were identified: Regulation Functions, Production Functions, Carrier Functions and Information Functions. Analyses of data gathered from different experts through elicitation of expert opinion and participatory rural appraisal were done to determine the cumulative effects of different stressors to valued ecosystem components which ultimately affects the services of wetlands. Prediction was done by developing a predictive model using Bayesian Network Modelling and WEKA software (Figure Below)

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WEKA software was used as the data mining tool. Bayes Net classification technique was used to develop the predictive model of the cumulative effects of different stressors to wetland services. Bayes Net presents the probability concisely and comprehensively in a graphical manner. The output is the Directed Acyclic Graph (DAG) reflecting the nodes with edges and the Conditional Probability Table showing the probabilities [7].

The model was constructed using the Simple Estimator for estimating the conditional probability table. It estimates probabilities directly from data. The alpha is set to 0.5. Alpha is used for estimating the probability tables and can be interpreted as the initial count on each value. K2 was used as the learning algorithm using Naïve Bayes as the initial network used for structure learning. Markov blanket was set to True to ensure that every attribute in the data represents the class attribute. Akaike Information Criterion (AIC) was used to evaluate the quality of the network structure. 10-fold cross validation was used.

(12) Results and Discussions

Given all variables observables, the directed acyclic graph was constructed to show the correlations among the stressors and the given valued ecosystem components. A CPT for each variable is generated to reflect the probabilities for each node given the occurrence of another node. Each variable is conditionally independent of its nondescendant in the network graph, thus the equation for joint probability distribution

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(Y_i))$$

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Hypothetically, land use change was identified to influence the other stressors as well as the ecosystem values. With the change in land use and the development scenarios, this will lead to a potential impacts on ecosystem values.

Impact of Stressors to Regulation Function

Based on the validation summary 216 or 97.30% of the instances are correctly classified and 6 or 2.70% of the instances are incorrectly classified. The confusion matrix reveals that 156 instances is classified as high, 44 as moderate and 1 instances is classified as low. There are 1 instance incorrectly classified as high, 4 incorrectly classified as moderate and 1 instance incorrectly classified low. The RMSE (standard deviation) is 0.1109 meaning the model is able to correctly predict the target value.

The directed acyclic graph shows that the direct primary stressors have great impact on the regulation function with the probability for high effect of 0.705, moderate effect of 0.217 and low effect of 0.078

The conditional probability equation of the regulation function is $P(RG)=P(RG|PL, LU, IS, OE, CC, AC, HP, MN, ID, AG)$

read as: $P(RG)$ given the occurrence of primary direct stressors (PL – Pollution, LU – Land Use, IS – Invasive Species, OE-Over Exploitation, CC – Climate Change, AC – Aquaculture, HP – Hydropower, MN-Mining, ID - Industries, AG - Agriculture, RG- Regulation)

Correctly classified instances	216 or 97.30%
Incorrectly classified instances	6 or 2.70%
RMSE	0.1109

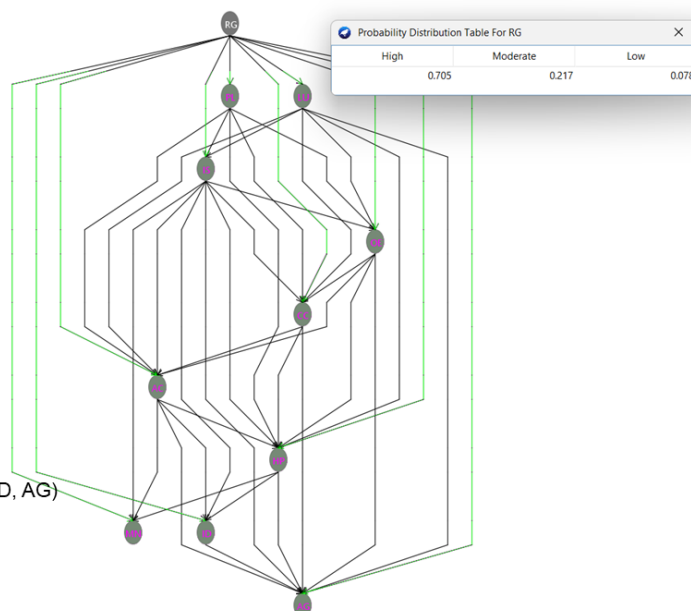
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=== Confusion Matrix ===
  a  b  c  <-- classified as
156  1  0 | a = High
  4 44  0 | b = Moderate
  0  1 16 | c = Low

```

Probability Equation:

$$P(RG)=P(RG|PL, LU, IS, OE, CC, AC, HP, MN, ID, AG)$$



Impacts of Stressors to Production Function

The cross-validation summary shows that 221 or 99.55% of the instances are correctly classified and 1 or 0.45% of the instances are incorrectly classified. The confusion matrix

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shows that 1 instance is incorrectly classified to high. An RMSE of 0.0488 means an almost best fit model.

The directed acyclic graphs shows that direct stressors affects the production function of wetlands. The conditional probability table for reflect the probability for low effect of 0.025, moderate effect of 0.195 and high effect of 0.781. The direct stressors have a high impact on the production function of wetlands.

The conditional probability equation of the production function is $P(PP)=P(PP|PL, LU, IS, OE, CC, AC, HP, MN, ID, AG)$

read as: $P(PP)$ given the occurrence of primary direct stressors (PL – Pollution, LU – Land Use, IS – Invasive Species, OE-Over Exploitation, CC – Climate Change, AC – Aquaculture, HP – Hydropower, MN-Mining, ID - Industries, AG - Agriculture, RG- Regulation)

Correctly classified instances	221 or 99.55%
Incorrectly classified instances	1 or 0.45%
RMSE	0.0488

=== Confusion Matrix ===

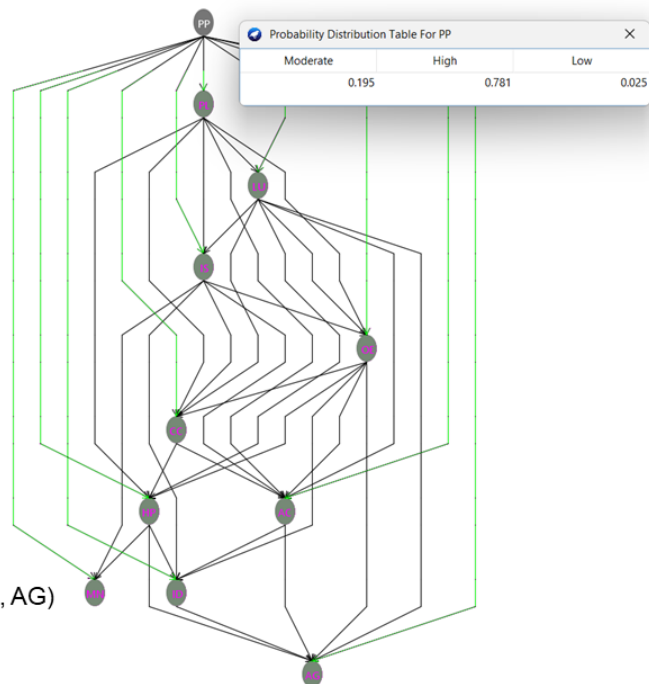
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a  b  c  <-- classified as
43  0  0 | a = Moderate
0 173  1 | b = High
0  0  5 | c = Low

```

Probability Equation:

$$P(PP)=P(PP|PL, LU, IS, OE, CC, AC, HP, MN, ID, AG)$$



Impact of Stressors to Carrier Function

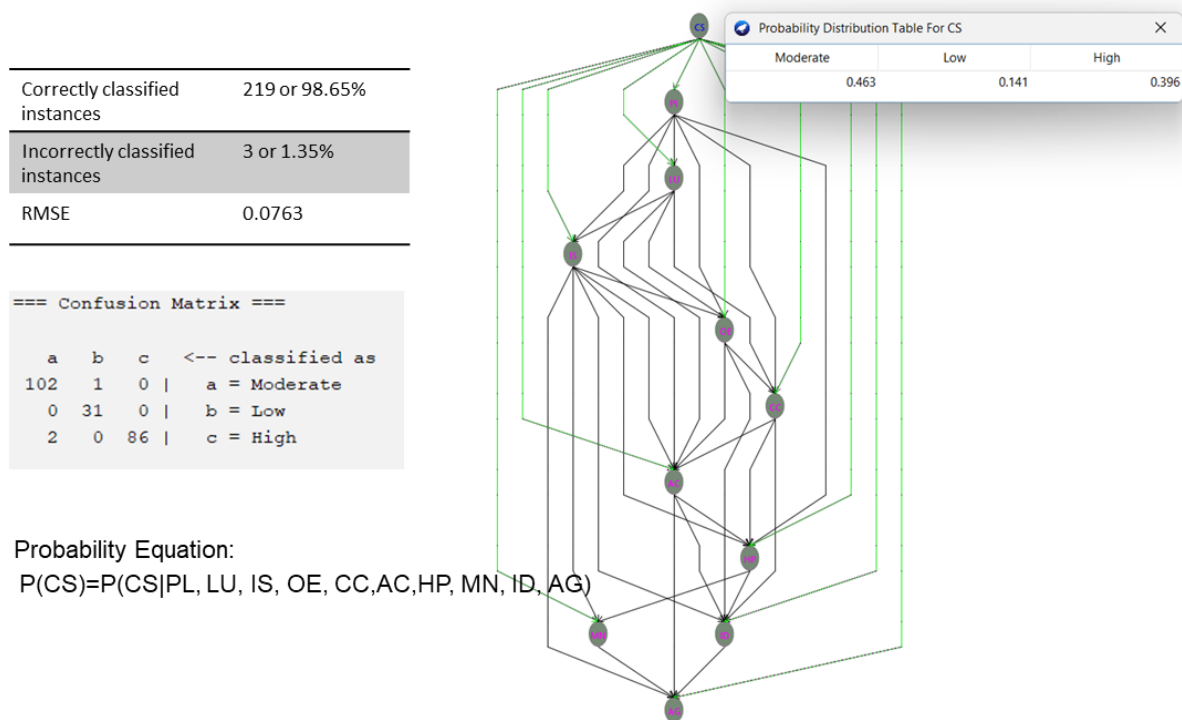
From cross-validation summary, 219 or 98.65% of the instances are correctly classified and 3 or 1.35% of the instances are incorrectly classified. 1 instance is incorrectly classified moderate and 2 instances incorrectly classified to high. An RMSE of 0.0763 means that data is around the line of best fit.

The directed acyclic graph shows the direct stressors that affects the carrier functions of wetlands. The direct stressors have a moderate impact on the carrier function of wetlands given the probability of 0.463. Moreover, the stressors have a probability of 0.396 and 0.141 for high effect and low impact on carrier services respectively.

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The conditional probability equation of the carrier function is $P(CS)=P(CS|PL, LU, IS, OE, CC,AC,HP, MN, ID, AG)$

read as: $P(CS)$ given the occurrence of primary direct stressors (PL – Pollution, LU – Land Use, IS – Invasive Species, OE-Over Exploitation, CC – Climate Change, AC – Aquaculture, HP – Hydropower, MN-Mining, ID - Industries, AG - Agriculture, RG- Regulation)



Impact of stressors to Information Function

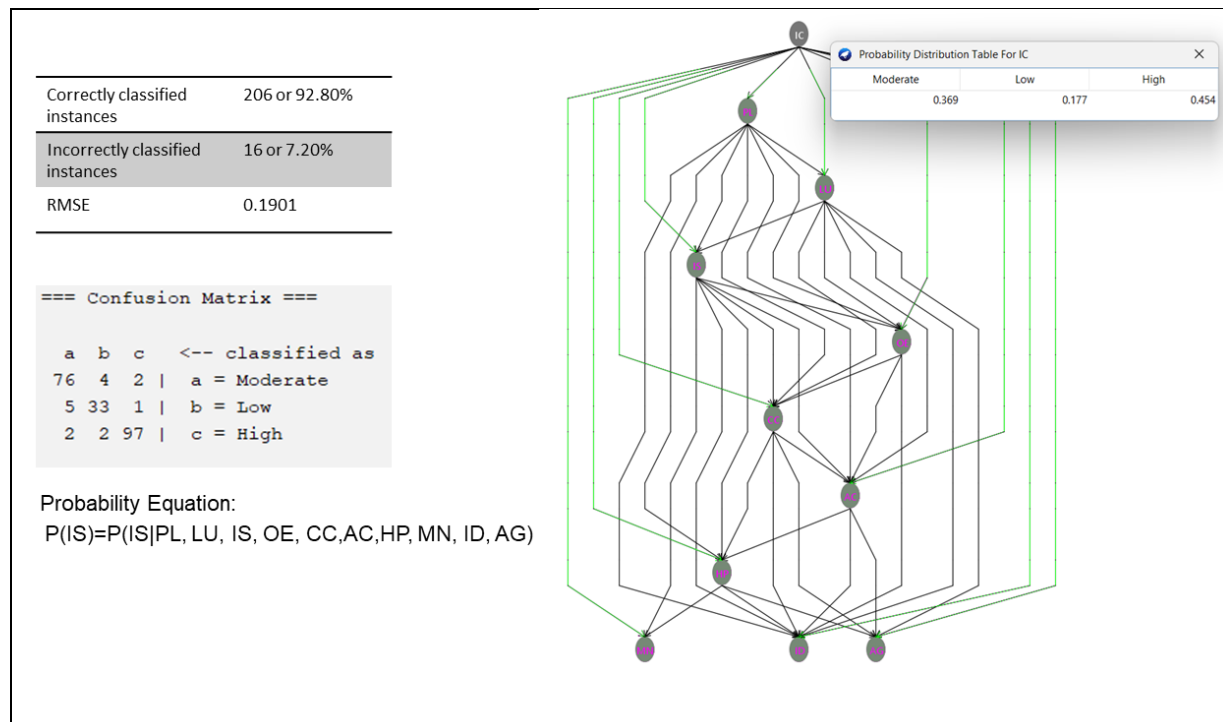
The cross-validation summary shows that 206 or 92.80% of the instances are correctly classified and 16 or 7.20% of the instances are incorrectly classified. An RMSE of 0.1901 means model is able to accurately classify the data.

The directed acyclic graphs shows that direct stressors affects the information function of wetlands. The conditional probability table for reflect the probability for low effect of 0.177, moderate effect of 0.369 and high effect of 0.454. The direct stressors have a high impact on the information function of wetlands.

The conditional probability equation of the Information function is $P(C-IS)=P(IS|PL, LU, IS, OE, CC,AC,HP, MN, ID, AG)$

read as: $P(IS)$ given the occurrence of primary direct stressors (PL – Pollution, LU – Land Use, IS – Invasive Species, OE-Over Exploitation, CC – Climate Change, AC – Aquaculture, HP – Hydropower, MN-Mining, ID - Industries, AG - Agriculture, RG- Regulation)

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(13) Conclusions and Recommendations

1. Multi-factorial analysis, intrinsic to CEA, can be addressed using Bayesian Network Modelling and was seen as a relevant approach to the wetland multifunction analysis.
2. The developed CEA methodological approach using Bayesian Network Modelling overcomes the difficulty of quantifying cumulative effects, thus providing a clear management prospective for wetlands.
3. The developed CEA methodological approach overcomes the enormous data complexity on interactions of numerous stressors and multiple functions.
4. Developed integrated quantitative CEA tool provided an innovative approach in the field of wetlands which stressed the significance of wetland ecosystems to human well-being which has been neglected.
5. The developed CEA tool can provide an effective management strategy which can balance development projects, wetlands integrity and human well-being.

It is therefore recommended to use the model in proper management of wetlands and to serve as a guide to policy makers.

(14) Output

Predictive Model for Wetland Management Strategy to balance economic development and ecological integrity.

(15) Technology Transfer Activity Conducted

(16) Economic Analysis

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4. Department of Sports, Culture and Tourism, Vietnam
5. Southeast Asian Fisheries Development Center (SEAFDEC, Philippines)
6. Department of Environment and Natural Resources(DENR, Philippines)

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